



SAFETY DATA SHEET

COPPER SULFATE

Aug 1 2020 (Jun 1 2023)

SECTION 1 - PRODUCT AND COMPANY IDENTIFICATION

Manufacturer: Stellar Solutions, Inc. (847) 854-2800
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EMERGENCY TELEPHONE: 1-800-535-5053 USA & Canada
24 hr INFOTRAC 1-352-323-3500 International

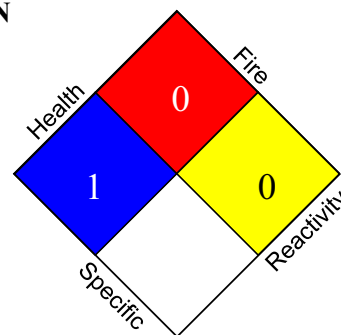
Trade Name: Copper Sulfate Passivation Test Solution
Product Use/Class: Standard test solution

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SECTION 2 - HAZARDS IDENTIFICATION

HEALTH	1
FLAMMABILITY	0
PHYSICAL HAZARD	0
PERSONAL PROTECTION	B

B: Safety glasses, gloves



*****EMERGENCY OVERVIEW*****: Harmful if swallowed. May cause eye burns or skin irritation.

Effects of Overexposure - Eye Contact: Liquid, aerosols and vapors of this product are irritating and can cause conjunctivitis, inflammation of the eyelid lining, ulceration, and clouding of the cornea. May cause permanent eye damage if not treated immediately.

Effects of Overexposure - Skin Contact: Prolonged or repeated exposures can irritate or burn skin. Reaction may be more severe if there is skin abrasion(s). Allergic sensitization in susceptible individuals is possible.

Effects of Overexposure - Inhalation: Inhalation is not an expected hazard unless misted or heated to high temperatures. Prolonged mist or vapor inhalation may cause irritation to the nose, throat, and upper respiratory tract.

Effects of Overexposure - Ingestion: Irritating to mouth, throat and stomach. May cause discomfort, nausea, vomiting, headache if swallowed. Large quantity may cause injury to the brain, stomach, liver, or kidneys.

Effects of Overexposure - Chronic Hazards: Repeated skin contact may cause a persistent irritation or dermatitis.

Primary Routes of Entry: Skin contact, skin absorption, inhalation, ingestion, eye contact

SECTION 2 - HAZARDS IDENTIFICATION (continued)**Hazard classifications:**

Acute toxicity, oral 4
 Skin corrosion/irritation 3
 Serious eye damage/eye irritation 2A
 Germ cell mutagenicity 2
 Carcinogenicity 1
 Reproductive toxicity 2
 Specific target organ toxicity, single exposure 1
 Hazardous to the aquatic environment, acute toxicity 1

Hazard statements: Danger

H302 Harmful if swallowed
 H316 Causes mild skin irritation
 H319 Causes serious eye irritation
 H341 Suspected of causing genetic defects
 H350 May cause cancer
 H361 Suspected of damaging fertility or the unborn child
 H370 Causes damage to organs
 H401 Toxic to aquatic life

**Precautionary statements:**

P260 Do not breathe dust/fume/gas/mist/vapours/spray.
 P273 Avoid release to the environment.
 P281 Use personal protective equipment as required.
 P301 + P312 If swallowed: Call a poison center or doctor/physician if you feel unwell.
 P302 + P352 If on skin: wash with plenty of soap and water.
 P305 + P351 + P338 If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
 P333 + P313 If skin irritation or rash occurs: Get medical advice/attention.
 P337 + P313 If eye irritation persists: Get medical advice/attention.

SECTION 3 - COMPOSITION/INFORMATION ON INGREDIENTS

<u>Item</u>	<u>Chemical Name</u>	<u>CAS NO.</u>	<u>% By Wt.</u>
01	Water	7732-18-5	97.37%
02	Copper Sulfate Pentahydrate	7758-99-8	1.58%
03	Sulfuric Acid	7664-93-9	1.05%

Exposure Limits:

<u>Item</u>	<u>TLV-TWA</u>	<u>TLV-STEL</u>	<u>PEL-TWA</u>	<u>PEL-CEILING</u>	<u>SKIN</u>
01	N.E.	N.E.	N.E.	N.E.	NO
02	N.E.	N.E.	N.E.	N.E.	NO
03	N.E.	N.E.	N.E.	N.E.	NO

(See Section 16 for abbreviation legend)

SECTION 4 - FIRST AID MEASURES

First Aid - Eye Contact: Remove contact lenses. IMMEDIATELY FLUSH EYES WITH WATER FOR AT LEAST 15 MINUTES, lifting lower and upper eyelids occasionally. Get medical attention immediately.

First Aid - Skin Contact: Wash with soap and water. Get medical attention if irritation develops or persists.

First Aid - Inhalation: Remove to fresh air. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Get immediate medical attention.

First Aid - Ingestion: If swallowed, induce vomiting. Give victim a glass of water or milk. Call a physician or poison control center immediately. Never give anything by mouth to an unconscious person.

SECTION 5 - FIRE AND EXPLOSION INFORMATION

Flash Point: N.A. - Aqueous solution

Flammable Limits: LEL: N.A. UEL: N.A.

Extinguishing Media: None required - Aqueous solution. Alcohol, foam, CO₂, dry chemical, water fog may be used.

Special Fire Fighting Procedures: If storage containers are involved in fire, keep cool with water spray to prevent pressure build-up. As in any fire, wear self-contained breathing apparatus pressure-demand (MSHA/NIOSH approved or equivalent) and full protective gear.

Unusual Fire and Explosion Hazard: None known. "Empty" containers retain some product residue (liquid and/or water vapor) and can be dangerous when pressurized. Bursting of containers can occur at elevated temperatures. Empty drums should be completely drained, properly bunged and promptly returned to a drum reconditioner, or properly disposed of.

SECTION 6 - ACCIDENTAL RELEASE MEASURES

Steps to be taken in case material is released or spilled: Absorb spill with inert material (e.g. dry sand or earth) then place in a chemical waste container. Avoid runoff into storm sewers and ditches which lead to waterways. Follow all government regulations.

SECTION 7 - HANDLING AND STORAGE

Handling: Wash thoroughly after handling. Do not get into eyes, on skin or on clothing.

Storage: Keep container closed when not in use. Store in corrosion resistant containers. Store away from incompatible materials. Store under a controlled environment. Avoid excessive heat. Keep from freezing.

SECTION 8 - EXPOSURE CONTROLS/PERSONAL PROTECTION

Engineering Controls: Good general ventilation should be sufficient to control airborne levels. Facilities storing or utilizing this material should be equipped with an eyewash facility and a safety shower.

Respiratory Protection: A respiratory protection program that meets OSHA 1910.134 and ANSI Z88.2 requirements must be followed whenever workplace condition warrant a respirator's use.

Skin Protection: Rubber, neoprene or polyvinyl chloride impervious gloves may provide protection against permeation. Gloves of other chemically resistant materials may not provide adequate protection. Consult your glove manufacturer for compatibilities.

Eye Protection: Wear chemical splash goggles. DO NOT WEAR CONTACT LENSES.

Other Protective Equipment: Wear suitable protective clothing to minimize and/or prevent contact. An eye-wash and safety shower should be present in the immediate area when handling this product.

Hygienic Practices: Wash hands before eating. Remove contaminated clothing and wash before reuse. Use only in well ventilated area. Follow all SDS/label precautions even after containers are emptied because they may retain product residues. Avoid contact with eyes, skin and clothing.

SECTION 9 - PHYSICAL AND CHEMICAL PROPERTIES

Physical State:	Liquid	Vapor Density:	Heavier than air
Appearance:	Blue	Vapor Pressure:	N.D.
Viscosity:	N.D.	Odor:	N.D.
Specific Gravity/Density:	1.01	Odor Threshold:	N.D.
pH @ 100 %:	< 2	Flammability:	N.A.
Solubility in Water:	Complete	Flash Point:	N.A.
Part. Coeff. n-octanol/water:	N.D.	Auto-ignition Temperature:	N.D.
Melting/Freezing Point:	32°F (0°C)	Decomposition Temperature:	N.D.
Boiling Point/Range:	> 212°F (100°C)	Upper Flammability/Explosive Limit:	N.A.
Evaporation Rate:	< 1 (Butyl Acetate = 1)	Lower Flammability/Explosive Limit:	N.A.

(See Section 16 for abbreviation legend)

SECTION 10 - STABILITY AND REACTIVITY

Stability: This product is stable under normal storage conditions.

Conditions to Avoid: N.A.

Incompatibility: Hydroxylamine, alkali metals, oxides of sulfur, strong oxidizers, strong reducing agents, caustic materials.

Hazardous Decomposition or Byproducts: N.A.

Hazardous Polymerization: Will not occur under normal conditions.

SECTION 11 - TOXICOLOGICAL PROPERTIES

Acute Toxicity	LD 50 (ORAL-RAT) 300 mg/kg (Copper Sulfate)
Threshold Limit Values:	LD 50 (ORAL-RAT) 2140 mg/kg (Sulfuric Acid) LC 50 (INHALATION-RAT) 510 mg/m ³ /2H (Sulfuric Acid)
Skin Corrosion/Irritation:	Causes mild skin irritation
Serious Eye Damage/Irritation:	Causes serious eye irritation
Respiratory Irritation:	Not expected to produce fumes causing irritation to the respiratory tract
Respiratory or Skin Sensitization:	Not expected to have sensitizing effects
Ingestion:	May cause gastrointestinal discomfort
Germ Cell Mutagenicity:	No known mutagenic effects
Carcinogenicity:	May cause cancer, genetic defects.
Reproductive Toxicity:	No known reproductive effects
Specific Target Organ Toxicity (STOT) single/repeated exposure:	No known effects on specific organs
Aspiration Hazard:	May cause irritation to the respiratory tract

SECTION 12 - ECOLOGICAL PROPERTIES

Ecological Information: Toxic to aquatic life. Avoid release to the environment. Dispose of contents in accordance with local, state, federal and international regulations.

SECTION 13 - DISPOSAL CONSIDERATIONS

Disposal Method: Follow all federal, state/province, and local regulations. Containers may be rinsed clean with water.

SECTION 14 - TRANSPORTATION INFORMATION

UN Number: Not listed

UN Proper Shipping Name: Not Regulated (Copper sulfate solution)

Marine Pollutant: Yes

DOT, TDG, IMDG, ICAO/IATA: Not Regulated for single or inner packaging of 5 L or less.

SECTION 15 - REGULATORY INFORMATION

U.S. Federal Regulations as Follows:

OSHA: Not listed.

CERCLA - SARA Hazard Category: This product has been reviewed according to the EPA "Hazard Categories" promulgated under Sections 311 and 312 of the Superfund Amendment and Reauthorization Act of 1986 (SARA Title III) and is considered, under applicable definitions, to meet the following categories:

Immediate Health Hazard

SARA Section 313: This product contains the following substances subject to the reporting requirements of Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 and 40 CFR Part 372:

Copper sulfate, sulfuric acid

Toxic Substances Control Act (TSCA): The chemical substances in this product are on the TSCA Section 8 Inventory.

This product contains the following chemical substances subject to the reporting requirements of TSCA 12(B) if exported from the United States:

No information is available.

Canada Regulations as Follows:

WHMIS: The Canadian Workplace Hazardous Materials Information System classification for this product is:

Copper sulfate, sulfuric acid

Refer elsewhere in the SDS for specific warnings and safe handling information. Refer to the employer's workplace education program.

SECTION 16 - OTHER INFORMATION**Date Prepared:** Feb 1 2016**Date Revised:** Aug 1 2020**Reapproved without revision:** Jun 1 2023**Previous Revision Date:** Mar 12 2018**Signature of Preparer:****Legend:**

N.A. - Not Applicable

N.E. - Not Established

N.D. - Not Determined

While the company believes that the data contained herein is factual and the opinions expressed are based on tests and data believed to be reliable, it is the user's responsibility to determine the safety, toxicity and suitability for their own use of the product described herein. Since the actual use by others is beyond our control, no guarantee, expressed or implied, is made by this company as to the effects of such use, the results to be obtained, or the safety and toxicity of the product, nor does this company assume any liability arising out of the use, by others, of the product referred to herein. Nor is the information herein to be construed as absolutely complete since additional information may be necessary or desirable when particular or exceptional conditions or circumstances exist or because of applicable laws or governmental regulations.